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Amphiphilic particulate phase semi-interpenetrating polymer networks based on recycled rubber matrix

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Abstract

The goal of this study is to enable the utilization of recycled rubber particles in aqueous media by producing amphiphilic particulate phase semi-interpenetrating polymer networks (PPSIPNs). Rubber granulates obtained from post-consumer rubber items were pulverized into fine particles using a solid state shear extrusion (SSSE) process. Reaction mixtures composed of toluene, acrylic acid (AA) monomer and azobisisobutyronitrile (AIBN) as an oil-soluble initiator were prepared and used to impregnate the produced rubber particles. Toluene was added as a co-swelling agent to induce more swelling of the rubber particles and, consequently, enhancement in the absorption of AA in the rubber network. The swollen particles were introduced into a micro-domain suspension polymerization reactor where the ionic strength of the aqueous phase was sufficient to prevent desorption of AA from the particles during the reaction. This resulted in formation and interpenetration of hydrophilic poly(acrylic acid) in the intermolecular structure of hydrophobic rubber network. The resulting composite particles are water dispersible and suitable for use in a variety of aqueous applications such as, additives to waterborne emulsions and vehicles for wastewater treatment.

Keywords: Amphiphilic; Interpenetrating polymer networks; Rubber recycling

1. Introduction

Recycling of scrap tire has become an environmental concern and a challenging technical problem. Over 283 million tires are scrapped every year in North America that nearly equates to 7 billion pounds of solid wastes [1]. The main problem with recycling of scrap tires is the crosslink structure of vulcanized rubber that prevents its melting and subsequent processing. To overcome this challenge, one approach is to revert the vulcanized rubber from a thermoset to a near thermoplastic using a devulcanization technique. The devulcanized rubber is suitable for reforming into new rubber items, usually accomplished by revulcanization. Alternative devulcanization techniques include ultrasonic [2], microwave [3], and chemical [4] treatments. A more feasible approach to rubber recycling is size reduction and pulverization of scrap tires to produce rubber particles [5]. Fine rubber particles can be incorporated into raw rubber or plastic formulations to form a new product. However, production of fine rubber particles is generally accomplished through the use of a cryogenic process that requires a high amount of cooling energy. Therefore, development of a pulverization technology to produce comparably fine rubber particles at ambient condition has been the subject of recent investigations [6–8].

Solid state shear extrusion (SSSE) is an ambient pulverization technology that is capable of producing fine rubber particles under high shear and compression forces [6,7]. Bilgili et al. [8] demonstrated that the SSSE process produces a rubber powder with higher surface area compared to other pulverization processes. They also observed a moderate level of degradation associated with the SSSE process and concluded that most chain breakages occur at weak sulfur bonds, resulting in particles with lower crosslink density [8]. Size reduction techniques and the SSSE process have been successfully applied to rubber recycling, but the applications for rubber particles have so

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far been limited to the use in non-aqueous media as non-reactive fillers [9,10].

Modification of recycled rubber particles to introduce chemical functionality or novel physical properties can significantly enhance their utility [11,12]. For example, addition of a crosslinkable functionality could allow the use of rubber particles as reactive fillers. Alternatively, modifications involving the addition of a hydrophilic character to the hydrophobic rubber can produce water dispersible particles and prevent their agglomeration in aqueous media [13,14]. This can provide a much broader range of application for recycled rubber particles, such as additives to waterborne polymeric coatings or in waste-water treatment. Synthesis of particulate phase semi-interpenetrating polymer networks (PPSIPNs) on a matrix of recycled rubber is one such modification approach, as presented in this work.

Semi-interpenetrating polymer networks (SIPNs) are polymeric composites obtained by interpenetration of a linear or branched polymer within a network of another crosslinked polymer [15]. SIPNs in particulate form (PPSIPNs) are more suitable for further processing and could have a variety of applications [16–18]. Amphiphilic SIPNs are produced when one of the polymeric components is hydrophobic and the other is hydrophilic [15,19]. In this study, amphiphilic PPSIPNs were prepared on a matrix of hydrophobic recycled rubber particles using poly(acrylic acid) (PAA) as the hydrophilic polymer.

There have been many studies focused on synthesis and morphology of PPSIPNs using the seeded emulsion polymerization method [20–24]. The mechanical and physical properties of the PPSIPNs depend on their morphology and dual phase continuity [15]. Complex morphologies could be obtained including core-shell or cellular structures based on the relative hydrophilicity of the polymers [21,22], swellability of the first polymer network by the monomer of second polymer [23], degree of crosslinking, concentration of initiator, polymerization reaction temperature [20,21] and degree of grafting of the second polymer [23]. However, addition of a good solvent as a co-swelling agent in the reaction mixture could significantly affect all these parameters [25,26]. Addition of a good solvent results in swelling of the first polymer network and enhancement in absorption of monomer of the second polymer [26,27].

In this study, toluene was used as an inert co-swelling agent along with AA monomer in the reaction mixture to enhance the diffusion of AA into the rubber networks. Interestingly, the amount of toluene used was found to have a significant impact on the morphology of the prepared PPSIPNs. Thermal analysis can be used to study the morphology of the produced PPSIPNs [28]. Our goal is to produce amphiphilic PPSIPNs on a matrix of recycled rubber particles to enable their use in a variety of aqueous media applications.

2. Experimental

2.1. Materials

Waste rubber slabs with approximate weight percent composition of 53.9% SMR-20 (natural rubber), 26.9% SRF (carbon black), 10.8% aromatic oil and 8.4% curatives and additives were shredded using a lab-scale Cumberland grinder. The shredded rubber granulates of 2–6 mm were pulverized into particles of different sizes using the SSSE pulverization process [7,8]. A narrow size fraction of the pulverized rubber particles in the size range of 250–420 μm were collected using sieving. The collected particles were repeatedly washed with toluene to remove all dirt and soluble additives. An inhibitor-removing column (Aldrich) was used to eliminate methyl ethyl hydroquinone (MEHQ) inhibitor from the acrylic acid monomer (99% AA inhibited with 200 ppm MEHQ, Aldrich). Azobisisobutyronitrile (AIBN 98%, Aldrich) initiator was used as received without further purification. The water used in all experiments was double distilled using Barnstead distillation columns.

2.2. Preparation of amphiphilic PPSIPNs

Ten swelling mixtures containing 10–100 vol.% AA in toluene were prepared as indicated in Table 1. All mixtures contained 0.3 mol% AIBN initiator on a solvent-free basis. Rubber particles in the amount of 0.5 g were soaked in the prepared de-aerated swelling mixtures. The slurries were allowed to equilibrate over night at room temperature and then centrifuged at 1000 rpm for 5 min to remove excess monomer/solvent. The centrifuged swollen particles were transferred into test tubes (10 ml Kimax brand glass test tube with screw cap) containing 1 molar sodium chloride (NaCl) solution. The prepared samples were degassed several times and blanketed with nitrogen gas to insure the complete removal of oxygen from the test tubes prior to sealing with a cap. The test tube samples were placed in a water bath orbital shaker (Barnstead Lab-line 3540) at an agitation speed of 200 rpm. The samples were kept in the

Table 1
Composition of the prepared reaction mixtures to impregnate the pulverized rubber particles

Sample	Volume ratio [AA]:[toluene] (ml)	Swelling coefficient
IP-0	0:10	2.75
IP-1	1:9	2.68
IP-2	2:8	2.54
IP-3	3:7	2.37
IP-4	4:6	1.96
IP-5	5:5	1.58
IP-6	6:4	1.23
IP-7	7:3	1.04
IP-8	8:2	0.79
IP-9	9:1	0.70
IP-10	10:0	0.54

shaker for 48 h at 65 °C for micro-domain suspension polymerization [29] of AA. The polymerized samples were washed with water to remove the salt and any PAA dissolved in the aqueous phase. The resulting particles were dried in a vacuum oven at 60 °C for 12 h to ensure the complete removal of the remaining solvent (toluene) and water.

2.3. Particle size distribution (PSD) measurements

The laser diffraction technique was used to measure the equivalent volume size distribution of the particulate materials using a Coulter LS 230 device. In this method, the suspended particles in a circulating fluid pass through a laser beam cell for size detection. The choice of mobile phase used in this measurement is critical as it may affect the dispersing characteristic of the particles [14]. Because the rubber particles are partially cross-linked, they do not dissolve in any solvent but they swell and disperse well in a good solvent like toluene. However, if toluene is used as the dispersing medium, the PSD measurement has to be corrected to obtain the size of unswollen particles. Assuming a spherical shape for the particles, this correction could be expressed by:

$$r_1 = r_2 \phi_2^{1/3} \tag{1}$$

where, ϕ_2 is the volume fraction of the rubber, r_2 and r_1 are the radii of the swollen and unswollen particles, respectively.

2.4. Swelling coefficient measurement

Approximately 0.5 g of the rubber particles was immersed in test tubes containing the swelling mixtures shown in Table 1. The samples were allowed to equilibrate at room temperature for a period of 24 h. The swollen particles were centrifuged at a speed of 1000 (rpm) for 5 (min) to remove the excess solvent and then immediately weighed. The equilibrium-swelling coefficient was calculated as:

$$\text{Swelling coefficient} = \left[\frac{W_2 - W_1}{W_1}\right] \tag{2}$$

where, W_1 and W_2 are the weights of dry and swollen rubber particles, respectively.

2.5. Concentration of monomer in the swollen particles

The concentration of monomer absorbed by the rubber particles was inferred from gas chromatography (GC) measurements on the remaining swelling mixtures using a Hewlett Packard 6890 spectrometer. The concentration of AA in the swelling medium was measured after swelling equilibrium state was reached and then subtracted from the concentration of AA in the initially prepared swelling mixture to calculate the amount of monomer absorbed by the rubber particles.

The kinetics of polymerization reaction was estimated by taking out a sample in intervals of 2 h and transferring it to an aluminum pan containing methanol and hydroquinone to stop the reaction. The fractional monomer conversion 'X_m' was calculated using gravimetric analysis:

$$X_m = \frac{p \times W_{IPN}}{M_0} \tag{3}$$

where, M_0 is the weight of monomer absorbed by the rubber particles; W_{IPN} is the weight of dried PPSIPN samples and p is the weight fraction of PAA in the composite particles.

2.6. Thermal characterization

The thermal degradation behavior of the prepared samples was studied using a thermal analyzer, Polymer Laboratories STA 625. Experiments were conducted in the temperature range of 35–550 °C under a nitrogen gas purge at scanning rate of 10 °C/min. Samples were prepared in open pans containing about 10 mg of the composite particles.

3. Results and discussion

3.1. Size distribution of the particulate materials

The PSD of the sieve-collected rubber particles in the range of 250–420 μm was measured in toluene using a laser diffraction particle size analyzer as shown in Fig. 1. As indicated in this figure and Table 2, the particle size measurement shows both smaller and larger particles than

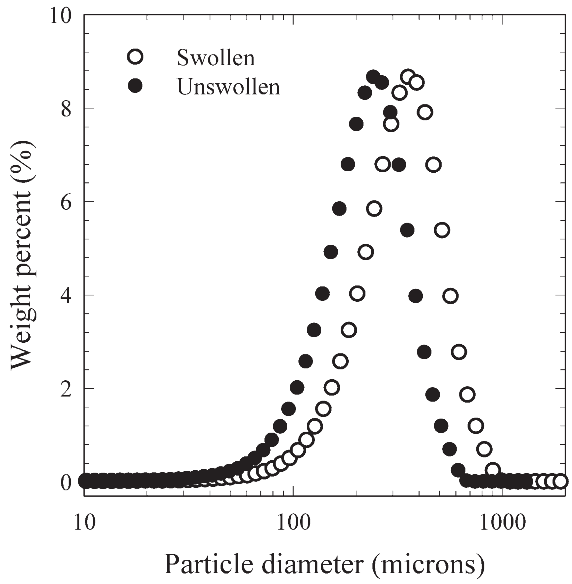

Fig. 1. Size distribution of the swollen rubber particles dispersed in toluene (open circles) using laser diffraction technique and the corrected size distribution for unswollen particles (solid circles).

Table 2
Size specifications of the particulate materials extracted from the size distribution measurements shown in Figs. 1 and 2

Sample	Mode (μm)	Median (μm)	Size range (μm)
Swollen particles	356	349	2–994
Unswollen particles	231	226	1–708
PPSIPNs (sample IP-5)	245	250	1–993

the ones measured using the sieves (250–420 μm). Smaller sizes were due to deagglomeration of the particles in toluene, which resulted from penetration of toluene in the voids between the primary particles. Larger sizes were due to the swelling of the individual particles. The volume fraction of rubber in the swollen particles was estimated as 0.29 using Eq. (1). Each size fraction of the swollen particles was corrected by this factor to obtain the size distribution of unswollen particles (see Fig. 1). As indicated in Table 2, more than 50% of the unswollen particles were smaller than the lower size limit (250 μm) of particle size measured using sieves. This seems to indicate that the particles produced using the SSSE pulverization process contain a large amount of agglomerates.

To determine water dispersibility of the prepared PPSIPNs and compare with unmodified rubber particles, the PSD of the samples was measured using water as the dispersing medium. The unmodified rubber particles formed agglomerates larger than the maximum size range of the instrument (2000 μm) confirming their poor dispersibility in water. On the contrary, the produced PPSIPNs are water dispersible and their PSD can be analyzed in this medium. The PSD of the produced PPSIPNs in water is shown in Fig. 2, and appears to be similar to the distribution of unmodified rubber particles in toluene. This could be explained by the fact that the produced PPSIPNs disperse better in water because of the formation of hydrophilic PAA on the surface of the hydrophobic rubber particles. However, a small amount of agglomerated PPSIPNs is observed on the right shoulder of their PSD between 700–1000 μm. These could be an indication that a small portion of the very fine particles in the initial sample does not end-up as PPSIPNs. These particles would naturally agglomerate in water and show up as the large size peak in Fig. 2. Shahidi et al. [14] investigated this phenomenon by dispersing the recycled rubber particles in different liquid media.

3.2. Monomer content of the swollen rubber particles

The swelling coefficient of the rubber particles in the prepared swelling mixtures was calculated using Eq. (2) as indicated in Table 1. The use of toluene in the reaction mixture was necessary despite the closeness of the solubility parameter of rubber, 19.8 $MPa^{1/2}$, and AA monomer, 24.6 $MPa^{1/2}$ [30]. As shown in Fig. 3, the swelling degree of the rubber particles decreases by increasing the volume fraction of AA in the reaction mixture. Addition of toluene resulted in swelling of the rubber networks and enhancement in absorption of AA. The amount of monomer absorbed by the rubber particles was measured using GC analysis.

The maximum monomer absorption was achieved in the case of reaction mixture initially contained 50 vol.% monomer as indicated in Fig. 4. The reason is that the reaction mixture was concentrated enough and the rubber particles were sufficiently swollen with toluene. Addition of monomer in the reaction mixtures containing less than 50 vol.% AA resulted in more absorption of AA due to extended swelling of the rubber particles in these samples.

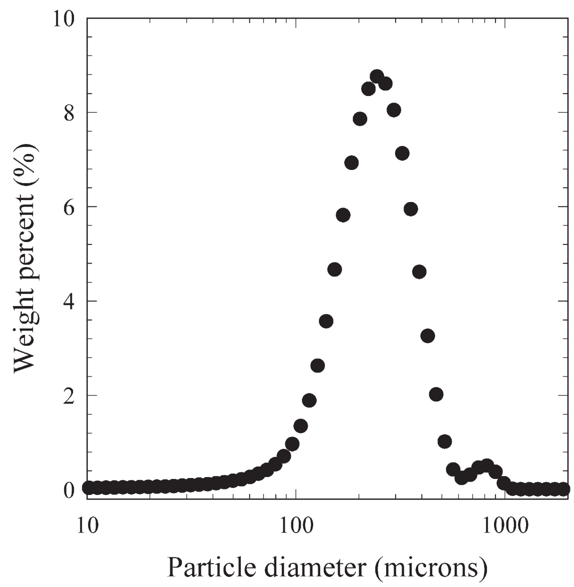

Fig. 2. Size distribution of the PPSIPNs (sample IP-5) dispersed in water.

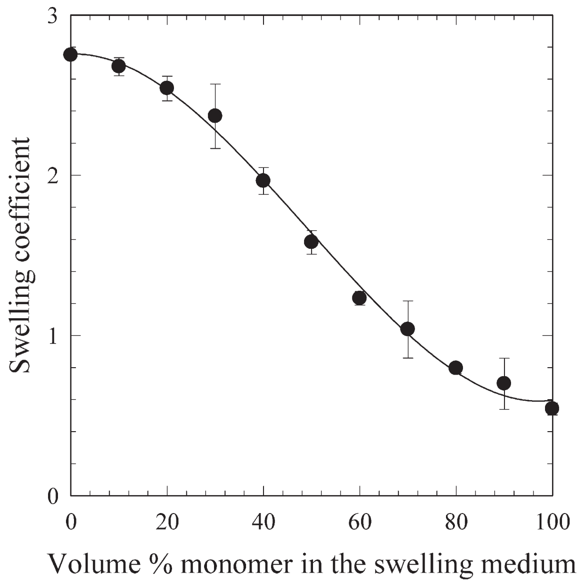

Fig. 3. Swelling degree of the rubber particles based on the volume fraction of AA monomer in the initially prepared reaction mixtures.

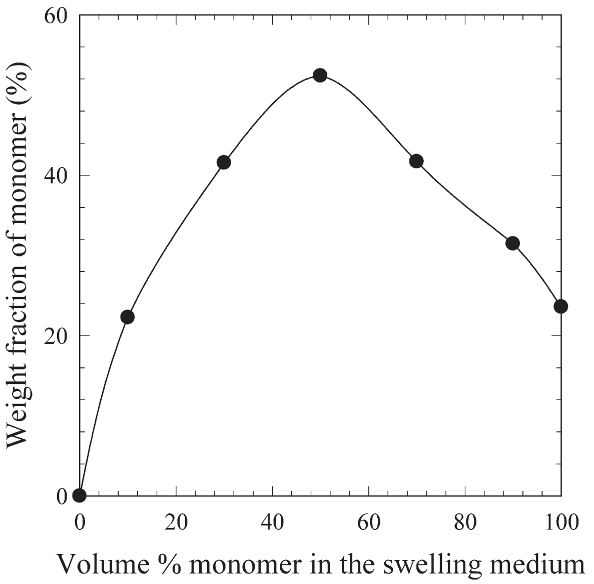

Fig. 4. Weight fraction of the monomer (AA) absorbed by the swollen rubber particles after the equilibrium swelling state was reached.

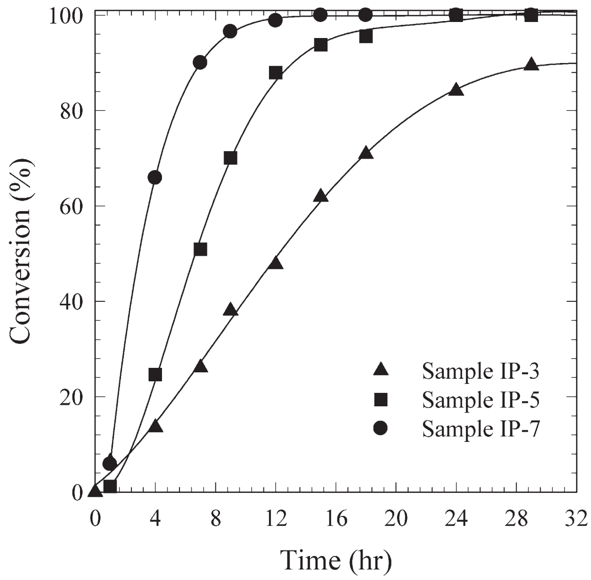

Fig. 5. Fractional monomer conversion obtained experimentally for the samples IP-3, IP-5 and IP-7.

Further addition of AA to more than 50 vol.% resulted in less absorption of monomer by the rubber particles. This is because of limited amount of toluene in these reaction mixtures and consequently, low swelling degree of the rubber particles.

3.3. Desorption of monomer into the aqueous phase

Polymerization reaction of AA was intended to take place within the hydrophobic phase (rubber particles) using an oil-soluble initiator (AIBN) and assuming that the AA monomer remains in the particles. However, the ionization of AA at the interface with water [31] results in desorption of AA into the aqueous phase of the system. By increasing the ionic strength of the aqueous phase and lowering the solubility of AA in the aqueous phase, desorption of monomer was controlled during the polymerization reaction. However, the polymerization reaction most likely occurred in the shell region of the particles and some desorbed in to the aqueous phase. At low ionic strength (e.g. 0.1 M NaCl), the PAA yield in the resulting particles was almost negligible in all samples, indicating excessive monomer desorption. This was also confirmed by pH measurement of the aqueous phase. The ionic strength was increased to 1 M NaCl electrolyte and subsequently desorption of the AA was lowered substantially.

3.4. Kinetics of polymerization and composition of the prepared PPSIPNs

The overall conversion of monomer was calculated and found to exceed 80% in all samples. Fig. 5 shows the kinetics of polymerization of AA in the selected samples IP-3, IP-5 and IP-7. The rate of polymerization reaction obtained experimentally was much lower than theoretically expected, which Lee et al. [32] have also reported. Most likely, this resulted from a non-uniform distribution of AA in the particles, as the hydrophilic monomer tends to migrate toward the aqueous interface. Thus, most of the polymerization reaction occurs in the shell region of the particles, where the available concentration of oil soluble initiator, AIBN, was lower than the total absorbed by the particles. In particular, due to a low concentration of the initiator in sample IP-3, the rate of polymerization was lower than the samples IP-5 and IP-7 (see Fig. 5). As a result, the viscosity of the polymerization loci within the swollen particles increased faster in the case of samples IP-5 and IP-7. It is expected that the PAA chains interpenetrated their molecular interstices within the rubber network in samples IP-5 and IP-7 more than sample IP-3. The study on the morphology of the produced particles was followed by the thermal degradation analysis of the samples.

Fig. 6 indicates the composition of prepared PPSIPN as a function of monomer volume fraction in the initially prepared swelling reaction mixture. The maximum PAA yield was achieved in the case of sample IP-5 that also absorbed the highest amount of monomer in the swelling step (see Fig. 4). Fig. 6 shows that there was a considerable difference between the total amount of PAA obtained in the test tube samples and the interpenetrated PAA in rubber particles. During the polymerization of AA, phase segregation occurs due to difference in hydrophilicity of PAA and rubber. Thus a large portion of PAA was desorbed into the aqueous phase.

3.5. Thermal characterization and morphology of the prepared samples

Fig. 7 shows the thermal behavior of prepared PPSIPN

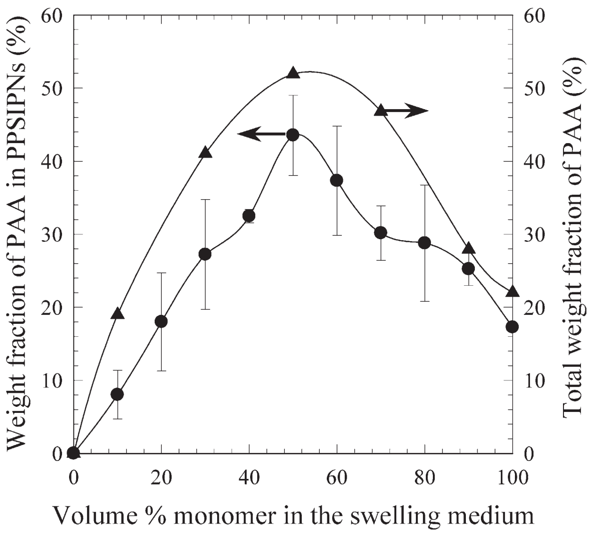

Fig. 6. Composition of the PAA yield in the rubber particles as PPSIPN with the total PAA including, PAA formed in the rubber particles, and dissolved in aqueous phase of the reaction.

samples as measured by differential scanning calorimetry (DSC). The DSC traces showed two endothermic peaks; one peak in the range of (150–280 °C) due to thermal degradation of PAA and the second peak in the range of (340–450 °C) that reflect the thermal degradation of rubber. The glass transition (T_g) temperature of rubber and PAA were individually measured as −60 °C and 110 °C, respectively. In a multi-phase component polymeric system, the T_g value of composite materials indicates the degree of miscibility [15]. Partial miscibility of the components results in a shift in T_g of each component toward the other

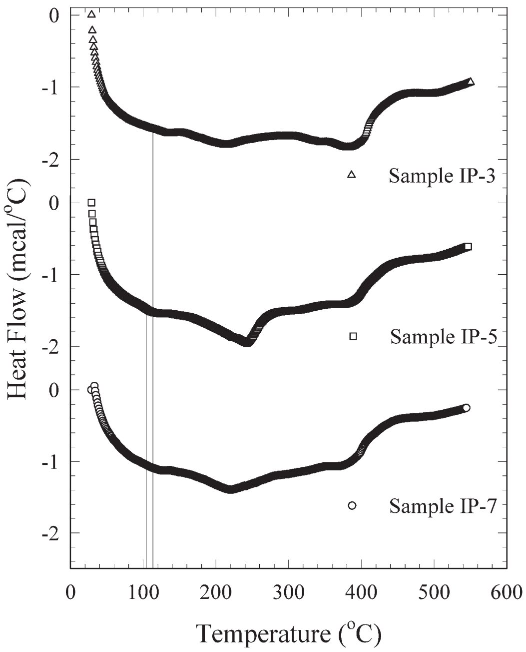

Fig. 7. Differential scanning calorimetry of the PPSIPN samples compared to the unmodified rubber particles (lines show the T_g of the PAA phase in the composite particles).

and a single T_g value could be obtained when the components are completely miscible [15]. The T_g value of PAA in the prepared PPSIPNs was measured from DSC results as indicated in Table 3. The T_g values of PAA in the case of samples IP-5 and IP-7 slightly shifted toward the T_g value of rubber (−60 °C), which indicates a partial interpenetration of PAA chains in the rubber network. To ensure the reproducibility of the results, these measurements were repeated several times and followed by thermogravimetric analysis (TGA) of the samples.

The weight loss of samples due to thermal degradation was obtained from the TGA results shown in Fig. 8. These results provide helpful information regarding the composition and phase structure of the samples. The TGA traces of samples in this figure show that two major weight loss events are observed in two distinct regions. The first weight loss was due to degradation of PAA (200–340 °C) and the second weight loss was due to degradation of rubber (340–460 °C). Carbon black and some other additives remain in the form of residues at the temperatures above 500 °C. Due to difference in the PAA content of the samples, the amount of residues at the end of experiments was different. Therefore, the residual content can be used to indicate the composition of prepared PPIPNs, knowing the amount of residues in the unmodified rubber particles is approximately 35 wt% (see Fig. 8).

As indicated in Fig. 8, the amount of residues was the same in the case of sample IP-3 and IP-7, which means they contain the same amount of PAA, as was also confirmed by the weight change measurement results shown in Fig. 6. However, different weight loss behavior due to degradation of PAA was observed in the temperature range of 200–340 °C. This could be explained by the fact that in sample IP-7, some parts of the PAA chains were deeply interpenetrated and trapped in the rubber network. Thus, the weight loss of this portion of PAA chains was observed during the degradation of rubber network in the temperature range of 340–460 °C.

Table 3 indicates the extent of degradation of sample components extracted from Fig. 9. The extent of degradation of unmodified rubber particles was measured as −0.8 wt%/°C. In the case of sample IP-3, the extent of degradation of rubber chains (DTG_{p2}) was similar to unmodified rubber sample. However, the DTG_{p2} value in the case of sample IP-5 and IP-7 was smaller than the unmodified rubber sample. These results indicated that more penetration of PAA within the rubber networks was

Table 3
Thermal characteristics of the produced PPSIPN composites

Sample	T_{g1} (°C)	DTG_{p1} (wt%/°C)	DTG_{p2} (wt%/°C)
IP-3	110	−0.06	−0.8
IP-5	105	−0.24	−0.64
IP-7	105	−0.16	−0.68

1 corresponds to the PAA phase and 2 relates to the rubber phase.

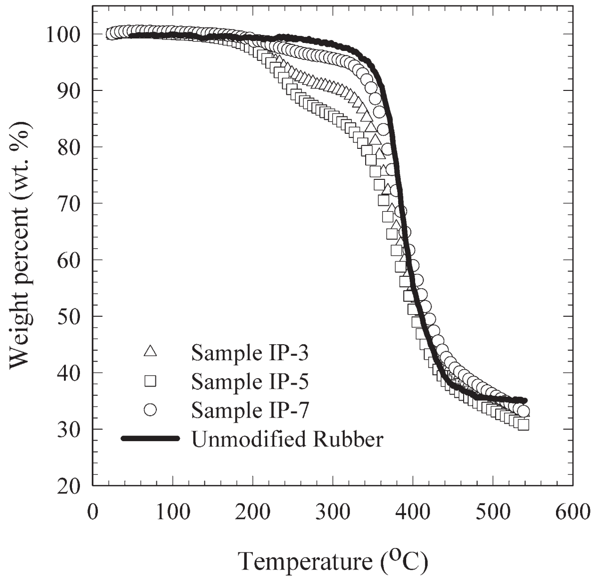

Fig. 8. Weight loss changes of the PPSIPN samples due to thermal degradation of rubber and PAA phases compared to the unmodified rubber particles.

achieved in samples IP-5 and IP-7 that improved the thermal degradation behavior of the rubber phase in these samples as shown in Fig. 9. As indicated in this figure, the degradation peak of the rubber phase in samples IP-5 and IP-7 shifted

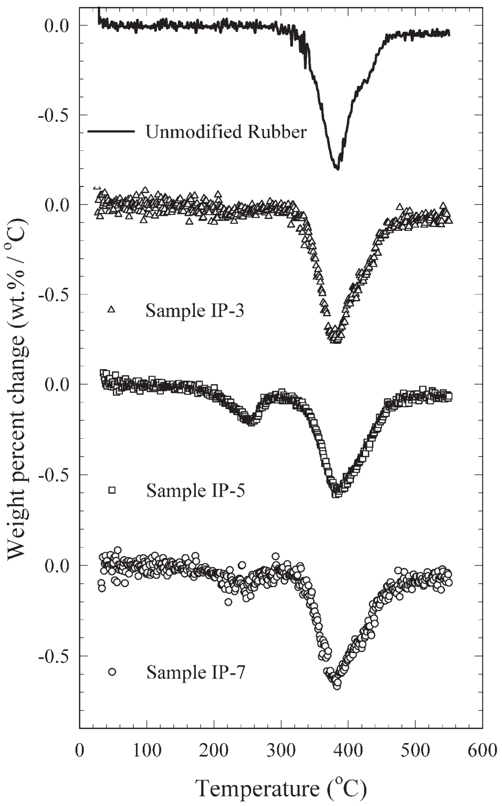

Fig. 9. Differential thermo-gravimetric analysis of the unmodified rubber particles compared to the PPSIPN samples.

toward higher temperatures compared to thermal degradation behavior of samples IP-3 and unmodified rubber.

4. Conclusions

Amphiphilic composite particles were successfully prepared on a matrix of recycled rubber material. Polymerization of monomer absorbed by the rubber particles resulted in partial interpenetration of PAA in the intermolecular structure of rubber networks. To increase the hydrophilicity of the rubber particles, the absorption of AA by the particles was enhanced by addition of toluene as a co-swelling agent. However, addition of toluene also resulted in less interpenetration of PAA in the rubber network. Therefore, by adjusting the amount of the solvent in the reaction mixture, the composition of the PPIPNs was better controlled. The resulting composite particles are water dispersible and suitable for use in aqueous media applications.

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